FAIRagro

Responsible Handling of Research Data

Legal and ethical aspects



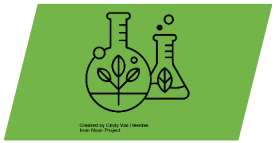
Copyright



CARE Principles



Data protection



Business-Secrets



Responsible Use of AI



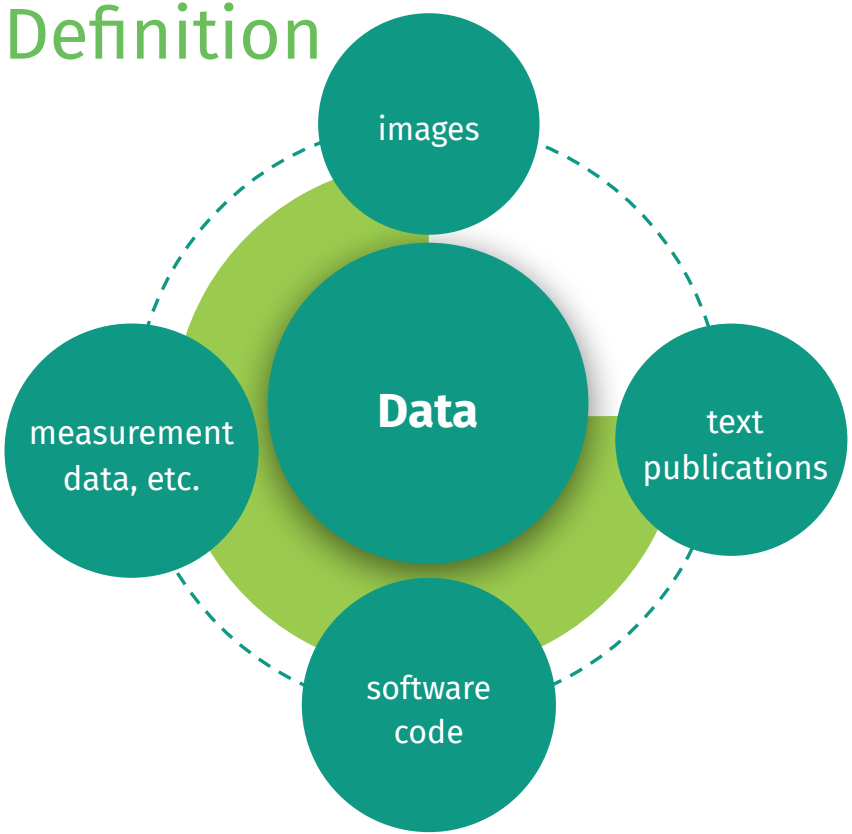
Nagoya-Protocol

Illustrations: “green chemistry”, “conservation”, “green energy”, “sustainability”, “gmo”, “wellness” by Cindy Van Heerden; “AI” by Merlin from Noun Project are licensed under CC BY 3.0

Data Rights – Copyright Law



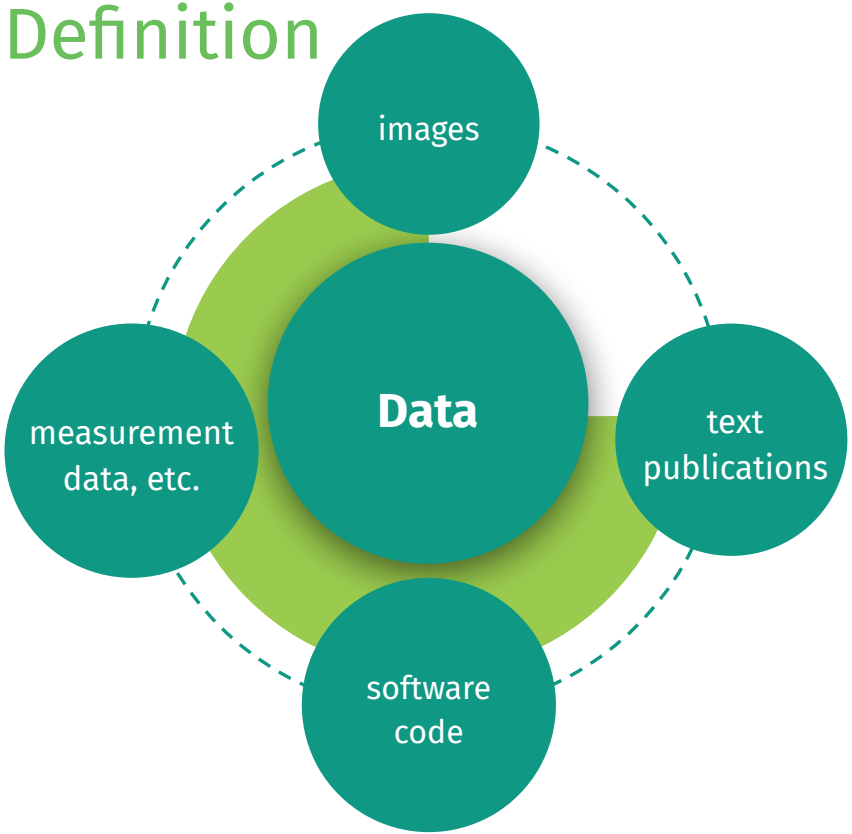
Data Rights Definition



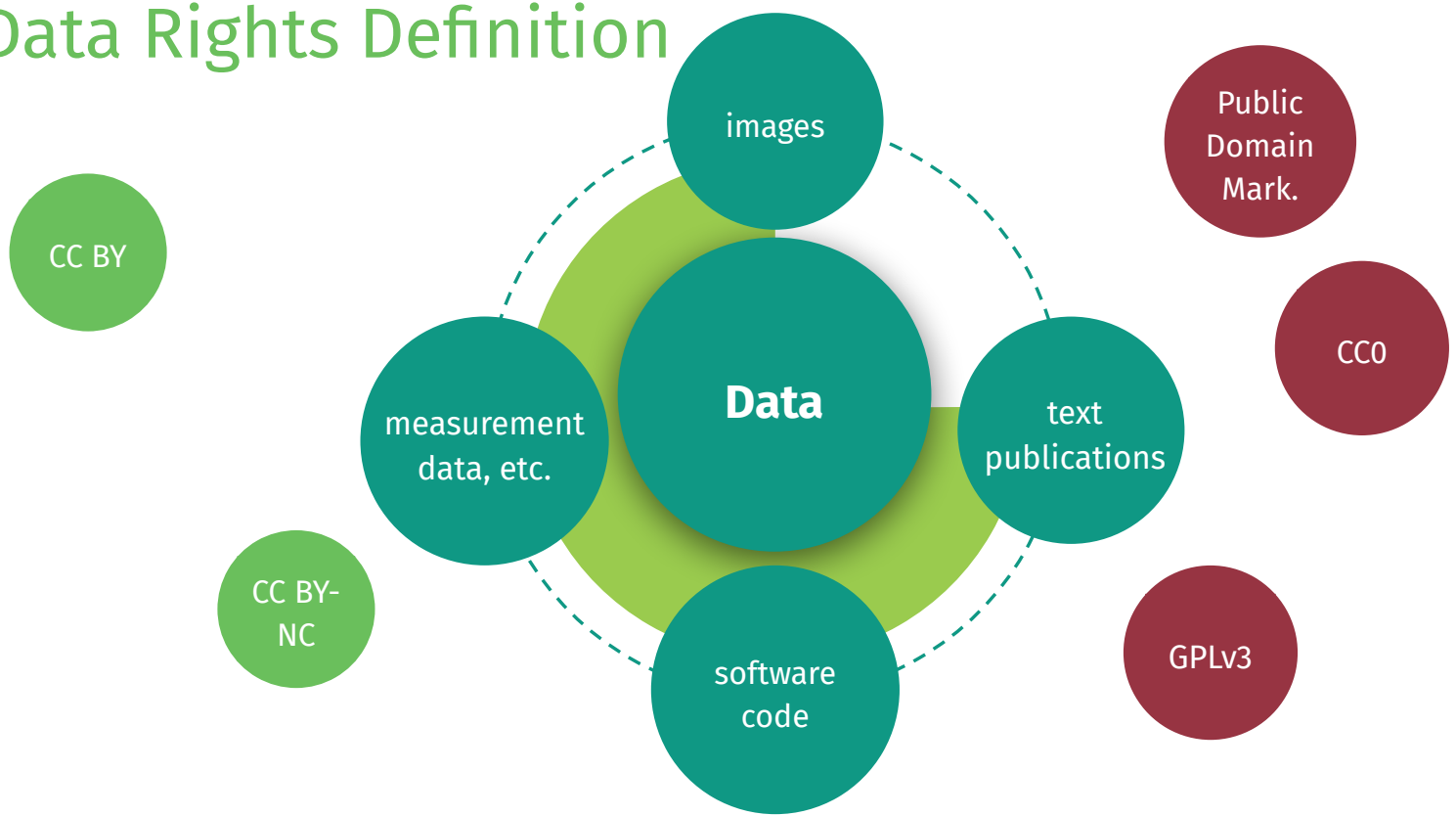
Data Rights Definition

possible protection:

- trademark law
- patent law
- copyright law

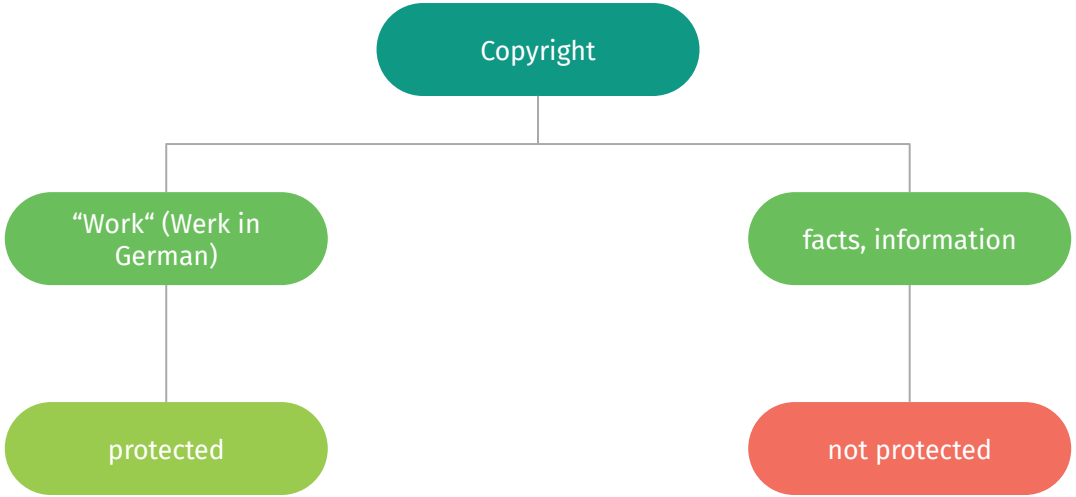


Data Rights Definition



Copyright Definition

Is it copyright protected?



Copyright Definition

Scope of originality

is there room to be creative?

-> not when strict technical specifications

+

Utilizing the scope in a creative manner

does the creator use that room in a creative way?

→

Threshold of originality

the creation is considered a “work” according to copyright law.

Copyright: Shape vs. Content

Protectable aka licenseable		Non-protectable
1	Research Paper (wording)	Findings presented in paper
2	Report on a case study (wording)	Data generated in Research
3	Software (program code)	Programming Language, algorithms, conceptual approach
4	Database (structure)	Content of databank

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Data Protection



Data that needs to be protected

- ➔ General Data Protection Regulation (GDPR): **personal data** - data that identifies natural persons
- ➔ Other “sensible data”:
  - Business Secrets
  - Location of endangered species



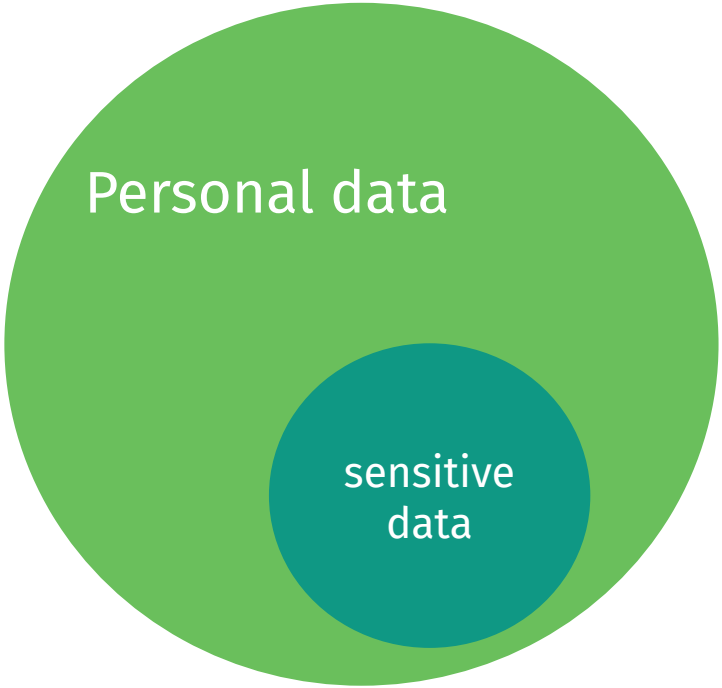
Created by jamzuri from Noun Project

Sources: GDPR, Lea Singson (2023); Illustration: “data” by jamzuri is licenced under CC BY 3.0

GDPR: personal vs. sensitive data

personal data

all information relating to **an identified or identifiable natural person**

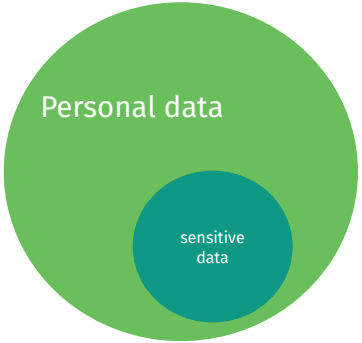


sensitive data

special categories of personal data that are subject to special processing conditions

Quelle: Europäische Kommission

GDPR: personal vs. sensitive data



sensitive data

in the GDPR-context: Art. 9

- data revealing racial or ethnic origin, political opinions, religious or philosophical beliefs
- **trade union** memberships
- **genetic**, **biometric** data, for the purpose of uniquely ident
  natural person
- **health data**
- data concerning a natural person's sex life

Important!
business/ contract secrets are not sensitive data according to the GDPR!

GDPR: identification vs. identifiability

identification:

➔ Identity results directly from the information itself

Bsp: Brigitte Meyer

identifiability:

➔ Identity results from linking several pieces of information

eg.: Assignment by means of identifier, identification number, location

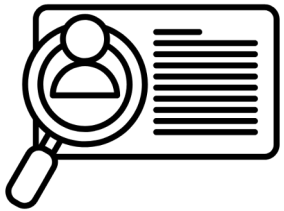


Illustration: “identify” by waqiahtul mukarromah is licensed under CC BY 3.0

GDPR: natural person vs. legal person

natural Person

- Brigitte Mayer, farmer

legal entity

- Association of persons that is vested with rights and obligations by law (legal entity)

➔ Apples & Pears Ltd.

GDPR: natural person vs. legal person

natural Person

- Brigitte Mayer, farmer

legal entity

- Association of persons that is vested with rights and obligations by law (legal entity)

➔ Apples & Pears Ltd.

Is there a difference
when it is the Brigitte Mayer ltd.?

Personal Data

What applies regarding agricultural data?

➔ personal data - data that identifies **natural persons, not companies!!**

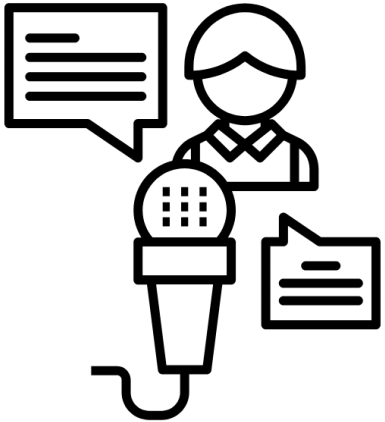


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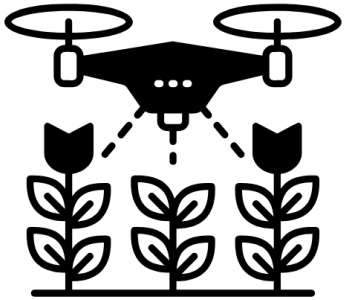
BUT Careful! agricultural farms are often not just companies, but homes of natural persons!
AND Careful! If name of the farm = name of the farmer

Sources: GDPR, Lea Singson (2023); Illustration: “data” by jamzuri is licenced under CC BY 3.0

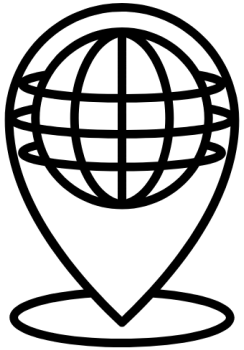
Personal Data in agricultural science



Created by seniman
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Created by Gendis Studio
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Illustrations: “drone” by Gendis Studio, “interview” by seniman, “geospacial technology” by Profit0101 from Noun Project is licensed under CC BY 3.0

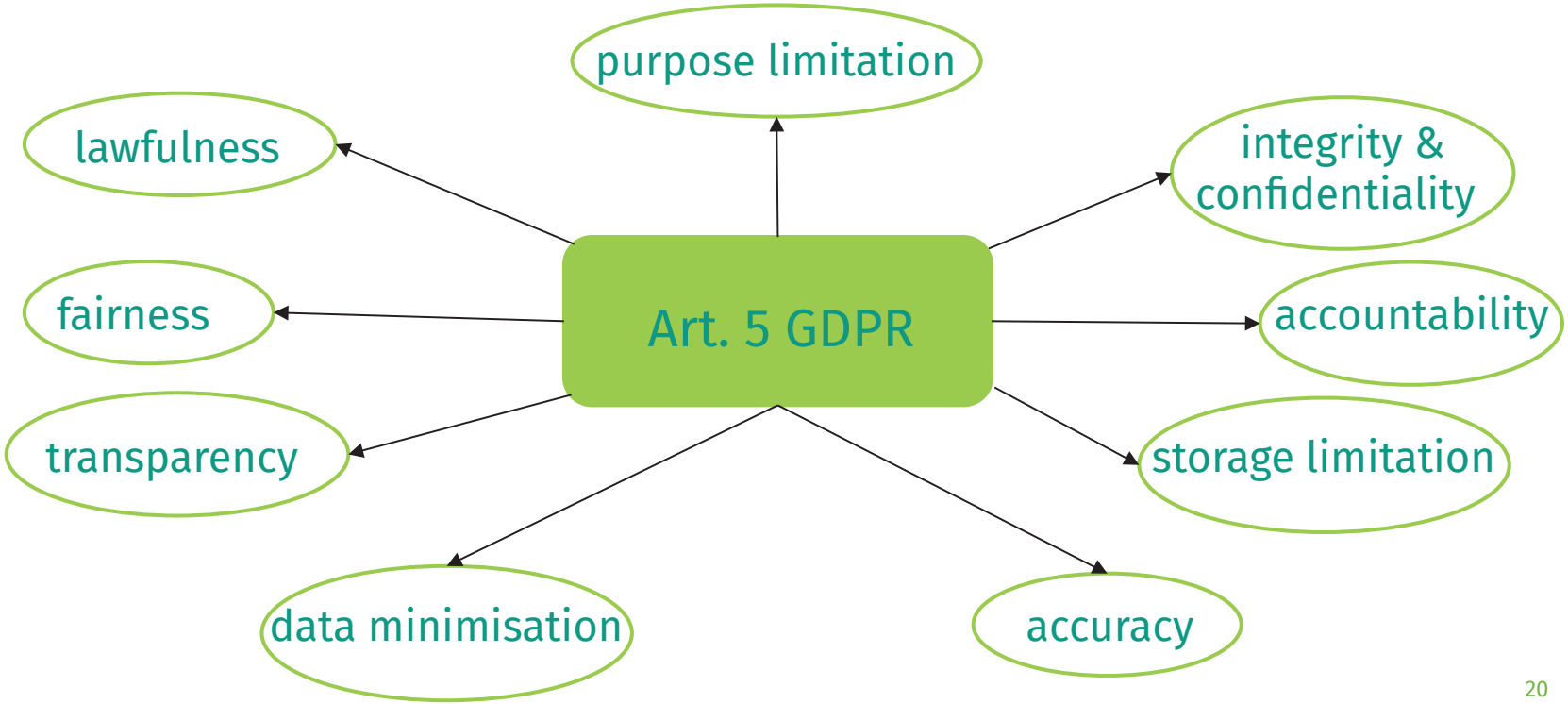
Data Protection - Applicability of GDPR

➔ please go to **menti.com**
➔ enter Code: **XXX XXX**
➔ or scan the QR Code



Assess the various scenarios according to whether the GDPR applies here.

When is data processing lawful?



Legal Basis of Data Processing (Art. 6 GDPR)

determines the legal grounds on which data processing can be built upon.

- ➔ Reminder: Principle of lawfulness (Art. 5 Abs. 1 a)
- ➔ Relevant for research i.p:

consent

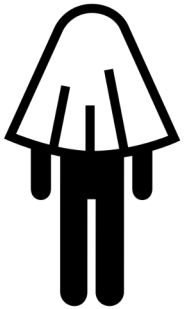
necessary for the performance of a contract

necessary for the performance of a task in the public interest

legitimate interest

no hierarchy!

Anonymisation



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from Noun Project

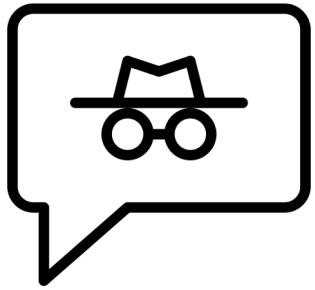
Anonymisation:
Modification of data with the result that it can no longer be linked to the personal information / only with unreasonable effort.

The GDPR does not apply

Illustration: “undercover” by Luis Prado from Noun Project is licensed under CC BY 3.0

Pseudonymisation

Pseudonymisation:
Tool for removing personal information from a data set. Direct or indirect identifiers are separated from personal data, stored separately and replaced by IDs.



Created by ismail abdurrasyid from Noun Project

GDPR applies!

Illustration: “pseudonym” by ismail abdurrasyid from Noun Project is licensed under CC BY 3.0

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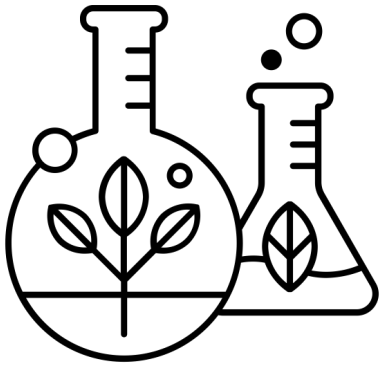
Business Secrets

Definition & Examples

Business Secrets (GeschGehG)

national law: “Geschäftsgeheimnisgesetz (GeschGehG)”

serves to protect business secrets from unauthorised acquisition, use and disclosure



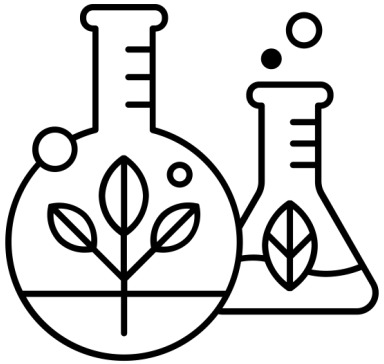
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Business Secrets (GeschGehG)

What is a business secret? (§ 2 Nr. 1 GeschGehG)
Information, that is

- of a technical or commercial nature
- not generally known/easily accessible to relevant parties
- therefore of commercial value
- protected by appropriate confidentiality measures
- subject to a legitimate interest in confidentiality



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Who is the owner of the business secret? (§ 2 Nr. 3 GeschGehG**)**
natural or legal person who has lawful control over the secret

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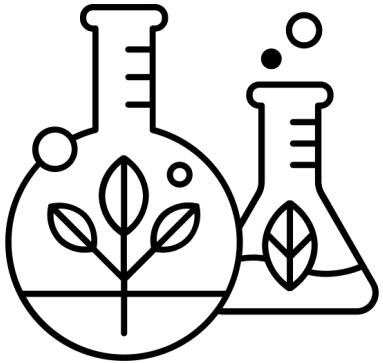
Business Secrets Examples

Manufacturing processes

cost information

business strategies

company data



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formulas and recipes

prototypes

market analyses

customer and supplier lists

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Nagoya Protokoll

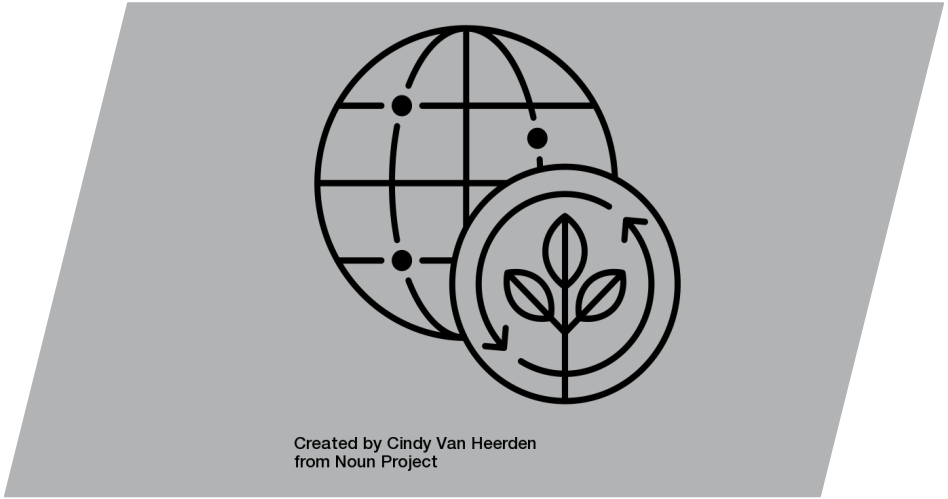
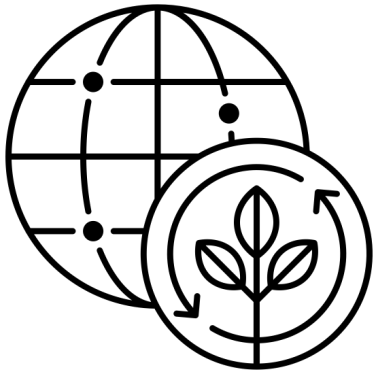


Abbildung: “sustainability” by Cindy Van Heerden from Noun Project is licensed under CC BY 3.0

Nagoya Protokoll

- Legal framework for the fair and equitable **sharing** of the **benefits** arising **from the utilisation** of **non-human genetic resources** amongst countries
- Access, Benefit-Sharing, Complience



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CARE-Principals

What exactly is it?

- The CARE Principles for Indigenous Data Governance were developed in 2019
- and are intended to complement the FAIR principles
- with a focus on the rights and interests of indigenous peoples

[Factsheet zu den CARE-Prinzipien]



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CARE-Prinzipien



Wann müssen die CARE-Prinzipien berücksichtigt werden?

- ➔ Betrifft ihre Forschung indigene Völker?
- ➔ Oder ihr kulturelles Erbe?
  - Sprache, Wissen, Praktiken, Technologien, natürliche Ressourcen
- ➔ Das kann auch ihr Territorium sein!
  - Flussbereiche, Waldgebiete, Gebirgsgebiete

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FAIRagro

AI Responsible Use

Approaching Research with AI

Do we need AI for this task?

- Is there available tests to evaluate the AI results?
- Does AI outperform simpler approaches?
- Do I need the largest general model for this task?
- Breaking down larger tasks into subtasks



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Results Evaluation

How to tell if the results and analysis are correct?

- **Sanity Checks**: Missing, inconsistent, implausible values
- **Automated Testing**: Compare against expected outputs
- **Manual Inspection**: Verify scientific plausibility



AI hallucinations and the importance of verification. Meme by unknown creator; reproduced in Dušek (2025, slide 10)

Responsible Use of AI in Research

Transparency

Disclose usage of AI - Report data biases

Verification

Asses the outputs

Data Protection

Watch out for data privacy and rights

Accountability

The researcher is responsible for all outputs

https://research-and-innovation.ec.europa.eu/news/all-research-and-innovation-news/era-living-guidelines-responsible-use-generative-ai-research-updated-2026-10-16_en